

VM177

Natural Frequency of a Submerged Ring

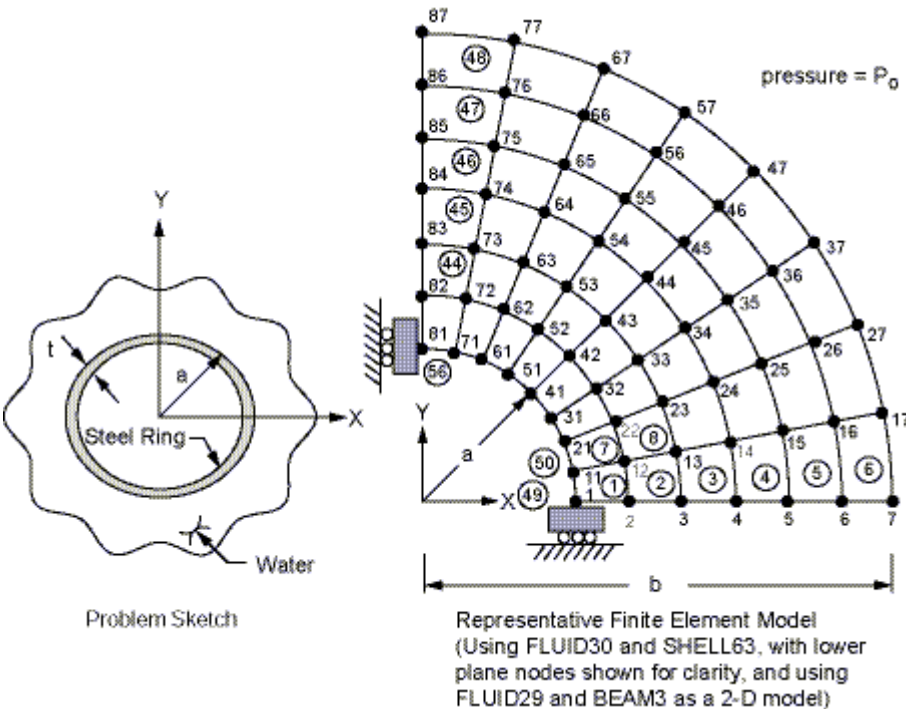
Overview

Reference:	E. A. Schroeder, M. S. Marcus, "Finite Element Solution of Fluid Structure Interaction Problems", Shock and Vibration Symposium, San Diego, CA, 1975.
Analysis Type(s):	Full Harmonic Analysis (ANTYPE = 3) Unsymmetric Matrix Modal Analysis (ANTYPE = 2)
Element Type(s):	3-D Acoustic Fluid Elements (FLUID30) Elastic Shell Elements (SHELL63) 2-D Acoustic Fluid Elements (FLUID29) 2-D Elastic Beam Elements (BEAM3) 4-Node Finite Strain Shell (SHELL181)
Input Listing:	vm177.dat

Test Case

A steel ring is submerged in a compressible fluid (water). Determine the lowest natural frequency for x-y plane bending modes of the fluid-structure system.

Figure 177.1 Submerged Ring Problem Sketch



Material Properties	Geometric Properties
For steel: $E = 30 \times 10^6$ psi $\nu = 0.3$ $\rho = 7.4167 \times 10^{-4}$ lb-sec ² /in ⁴	$a = 10$ in $b = 30$ in $t = .25$ in

For water:

C = 57480 in/sec

(speed of sound in water)

 $\rho = 9.6333 \times 10^{-5} \text{ lb-sec}^2/\text{in}^4$ **Analysis Assumptions and Modeling Notes**

For this problem, the fluid is assumed as extending only to a finite radius b where the pressure P_0 is zero and b is taken to be 30 inches. From the reference, this assumption should result in an error of less than 1% compared to the frequency for an unbounded fluid ($b = \infty$). The natural boundary conditions at the coordinate axes imply that $\delta P / \delta x = 0$ at $x = 0$ and $\delta P / \delta y = 0$ at $y = 0$.

This problem is solved using three separate analyses. The first, using 3-D acoustic fluid elements ([FLUID30](#)) with quadrilateral shell elements ([SHELL63](#)), the second, using 2-D acoustic fluid elements ([FLUID29](#)) with 2-D elastic beam elements ([BEAM3](#)) and the third using 3-D acoustic fluid elements ([FLUID30](#)) with quadrilateral shell elements ([SHELL181](#)).

In the first and third case, due to fluid-structure coupling involving unsymmetric matrices, the natural frequency is determined by performing a full harmonic ([ANTYPE](#) = 3) analysis with a frequency sweep. Monitoring the displacement of several key nodes (nodes 1, 41 and, 81) over the frequency range indicates the approximate desired natural frequency as the point where the nodal displacements indicate a resonant condition.

In the second case, the lowest natural frequency is obtained using an unsymmetric matrix modal analysis ([ANTYPE](#) = 2) corresponding to the frequency of mode one of the frequency data from the Lanczos unsymmetric eigensolver.

A preliminary finite element analysis (not shown here) was used to estimate a narrow range in which the 1st even bending mode frequency occurs. In the first case, two unit loads at nodes 1 and 81 are arbitrarily used to excite the desired bending mode of vibration.

Results Comparison

		Target ^[1]	ANSYS	Ratio
FLUID30 and SHELL63	f_1 , Hz	35.62	$36.61 < f < 36.62$	1.028
FLUID29 and BEAM3	f_1 , Hz	35.62	36.54	1.026
FLUID30 and SHELL181	f_1 , Hz	35.62	$36.61 < f < 36.62$	1.028

1. Solution from the reference under the same assumptions mentioned in the modeling notes.

Figure 177.2 Node 1 Displacement vs. Driving Frequency Near 1st Bending Mode Natural Frequency (Full Harmonic Analysis)

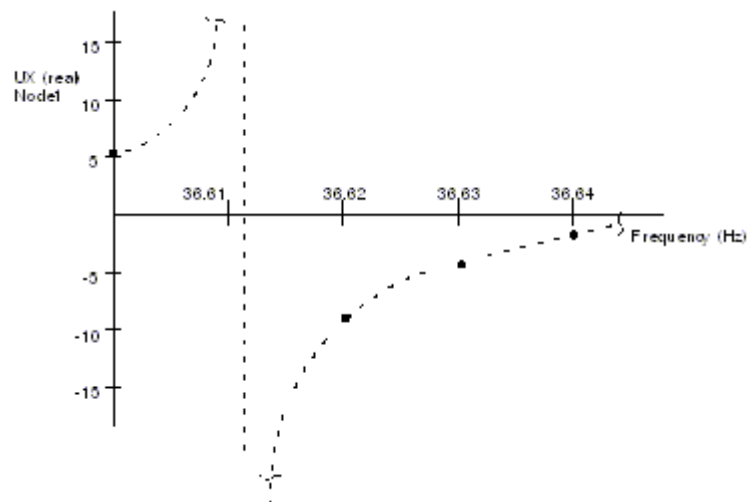


Figure 177.3 Real Displacement Component

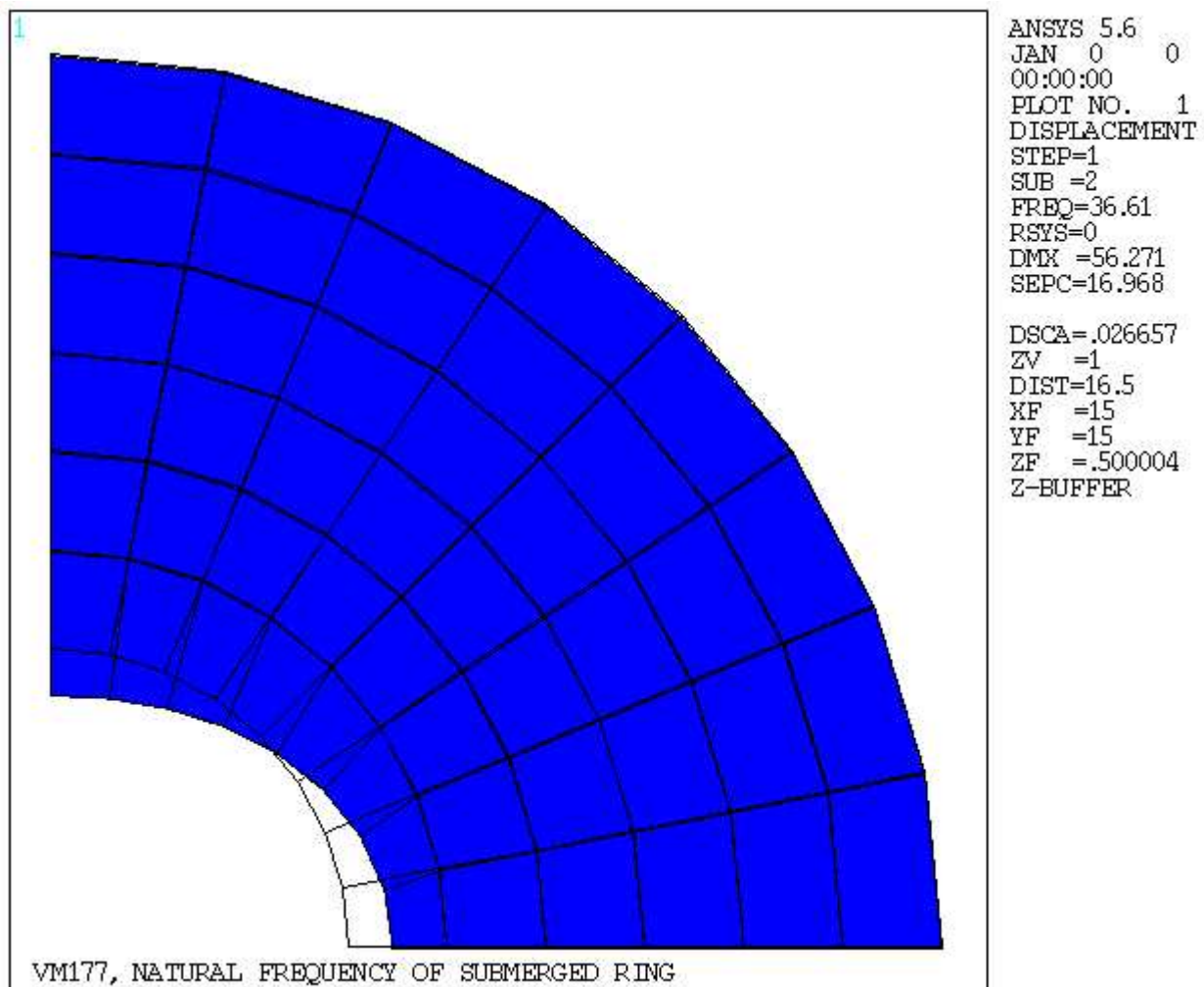
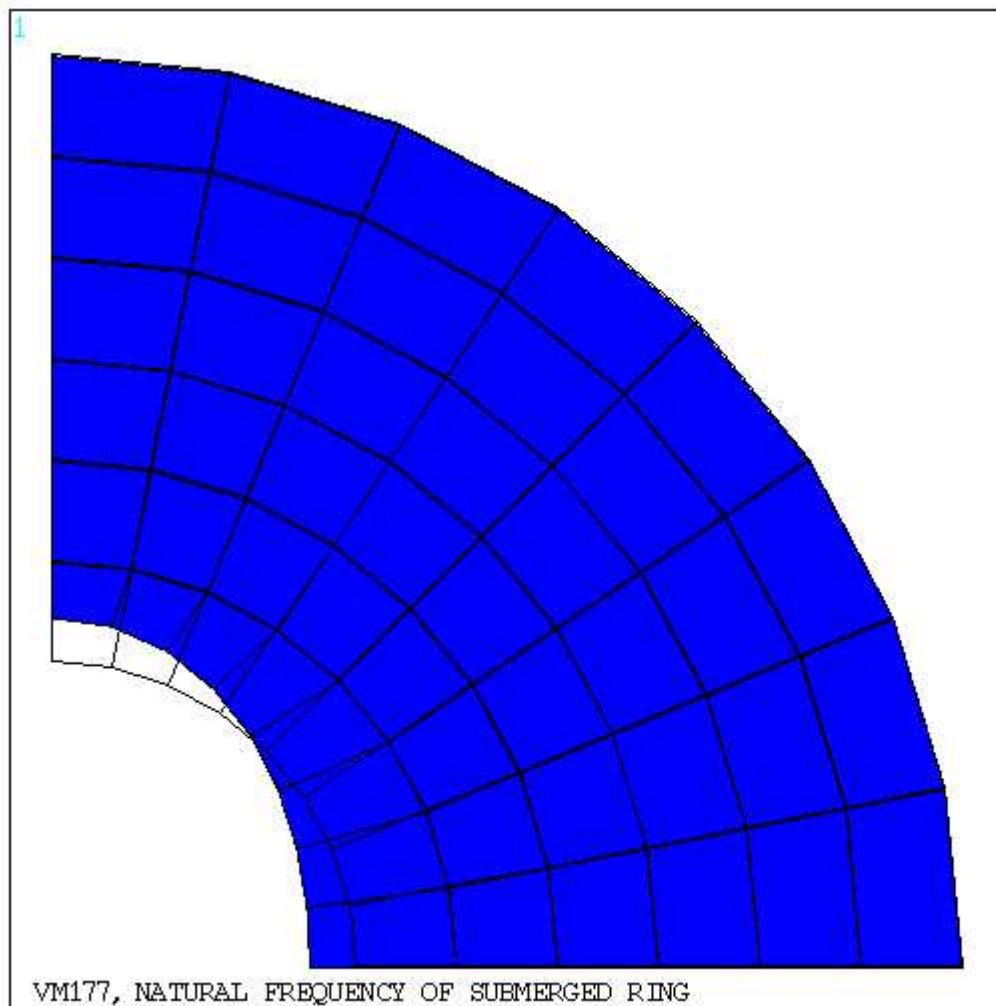


Figure 177.4 Real Displacement Component



ANSYS 5.6
JAN 0 0
00:00:00
PLOT NO. 2
DISPLACEMENT
STEP=1
SUB =2
FREQ=36.61
RSYS=0
DMX =56.271
SEPC=16.968

DSCA=.026657
ZV =1
DIST=16.5
XF =15
YF =15
ZF =.500004
Z-BUFFER